TITLE: Introduction to C and data types

Objective

1) To be familiar with structure of C-program.
2) To be familiar with printf() and scanf().
3) To be familiar with data types.
4) To be familiar with escape sequences.

PROGRAMS:

1) Write a program in C to print "Welcome to C-programming."

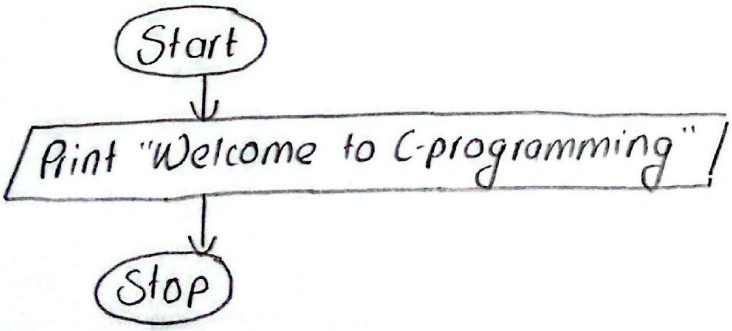
Algorithm:

Step1: start
Step2: print Welcome to C-programing
step3: stop.

C-programing:

```
#include <stdio.h>
int main()
{
print("Welcome to C");
return 0;
}
```

flowchart:



2) Write a program in C to illustrate use of \n and \t

Step 1: Start

Step 2: input P(principal) \t, T(time) \t, R (Rate)

Step 3: Calculate Simple interest: SI: (P*T*R) /100

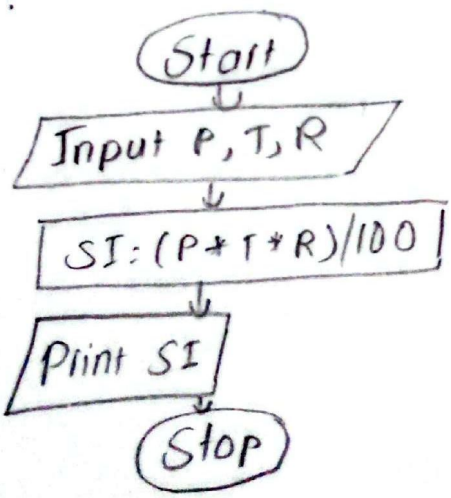
Step 4: Display SI

Step 5: stop

C-programme:

```
#include <stdio.h>
int main ()
{
flat P, T, R, SI;
printf("Enter principal \t ");
scanf ("%f", &P);
printf ("Enter time \t");
scanf("%f", &T);
printf ("Enter Rate \t");
scanf ("%f", &R);
SI: (P*T*R) /100;
printf ("\n Simple interest is: \t %2f", SI);
return 0;
}
```

flowchart:



3) Write a program in C to find sum of two numbers and display

Algorithm

Step1: start

Step 2: Declare two variables a and b

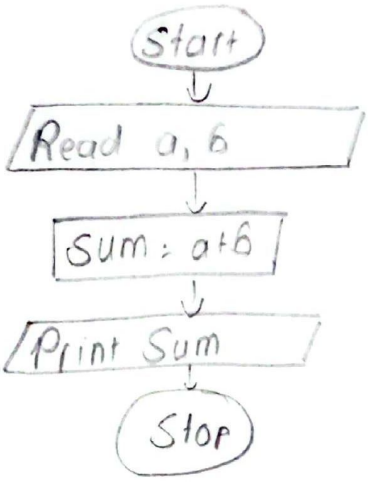
Step 3: Read the values of a and b

Step 4: Calculate sum = a+b

Step 5: Display the value of sum

Step 6: stop.

flowchart



Cprogramme:

```
#include <stdio.h>
int main()
{
int a, b;
printf("Enter the value of a: %d", a);
scanf("%d", &a);
printf("Enter the value of b: %d", b);
scanf("%d", &b);
c=a+b;
printf("The sum of %d and %d is", a, b);
return 0;
}
```

4) Write a program in C to swap numbers.

C-programme:

```
#include <stdio.h>
int main()
{
int a, b, temp;
printf("Enter first number:\n");
scanf("%d", &a);
printf("Enter second number:\n");
scanf("%d", &b);
temp=a;
a=b;
b=temp;
printf("\n After swapping:\n");
printf("\t first number= %d\n", a);
printf("\t Second number= %d\n", b);
return 0;
}
```

Algorithm:

Step1: start

Step2: Declare three variables a, b, and temp

Step3: Read values of a and b

Step 4: store a in temp

Step5: store b in a

Step 6: store temp in b

Step 7: Display the swapped values

Step 8: stop.

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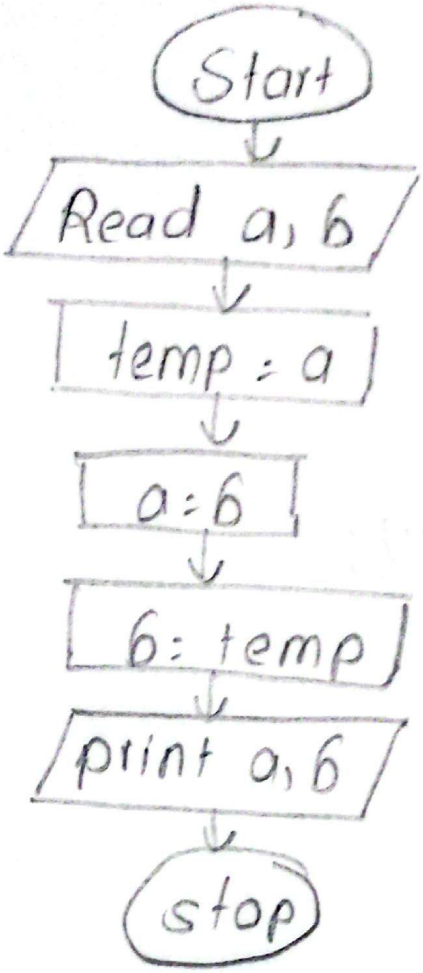
flowchart



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5)

Cprogramme.

```
#include <stdio.h>
int main()
{
int i;
float f;
char c;
char s[100];
printf("Enter integer:");
scanf("%d",&i);
printf("Enter float:");
scanf("%f",&f);
printf("Enter character");
fflush(stdin);
scanf("%c",&c);
printf("Enter string:");
scanf("%s",&s);
printf("Integer:%d, float:%0.2f, character: %c and string
= %s");
return 0;
}
```

6.

Algorithm

Step1: start

Step 2: Declare variables r and area

Step 3: Read the values of radius r

Step 4: calculate area: 3.14*r*r

Step 5: Display the value of area

Step 6: stop

Cprogram:

```
#include <stdio.h>
int main ()
{
float r, area;
printf ("Enter the radius :\n");
scanf ("%f", &r);
area = 3.14 * r * r;
printf ("\n Area of circle:\t %f \n", area);
return 0;
}
```

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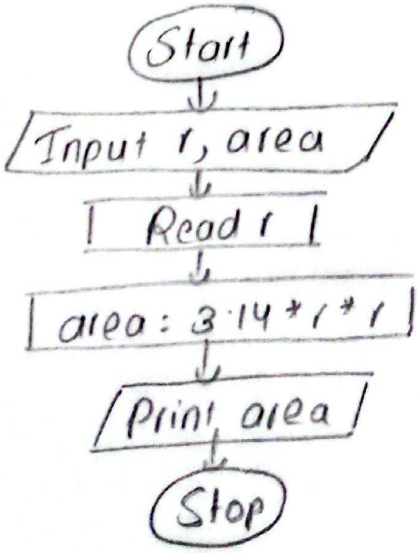

Conclusion:

In summary, the introduction to C helped us to understand the basics of C-programming.

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